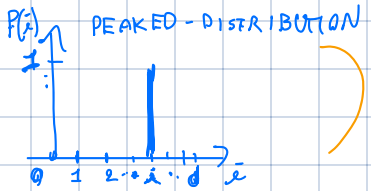
- $X = \{x_1, \dots, x_d\}$ Random Variable $\overset{\text{DEF.}}{\iff}$ $\begin{cases} \cdot\ p_i \geq 0 \\ \cdot\ \sum_{i=1}^{d} p_i = 1 \end{cases}$

  p_1 ↓ Probability to extract x_1 ; p_d ↓

- **DEF:** (SHANNON ENTROPY of X) $\quad H(X) := -\sum_{i=1}^{d} p_i \log(p_i)$

- $H(X) \geq 0$ $\quad \left(H(X) = 0 \iff \exists\, j \in \{1, \dots, d\} : p_i = \delta_{i,j} \right)$

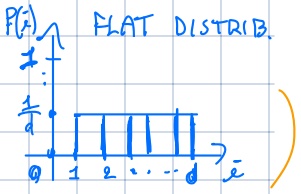


PROOF:

- $0 \leq p_i \leq 1 \Rightarrow -p_i \log(p_i) \geq 0 \Rightarrow H(X) \geq 0$
- $H(X) = 0 \Rightarrow p_i \log(p_i) = 0 \ \forall i$, with $p_i \geq 0$ and $\sum_{i=1}^{d} p_i = 1$

  $\Rightarrow p_i = 0$ or $\log(p_i) = 0$ ($\Leftrightarrow p_i = 1$) $\Rightarrow \exists\, j : p_i = \delta_{i,j} \ \forall i$.

- $H(X) \leq \log(d)$ $\quad \left(H(X) = \log(d) \iff p_i = \frac{1}{d} \ \forall i \right)$



PROOF:

- DEF: $f(x)$ is CONVEX (CONCAVE) $\iff f\left(\sum_{i=1}^{d} p_i x_i\right) \leq \sum_{i=1}^{d} p_i f(x_i)$ (≥) where $x_i \in \mathbb{R}$, $p_i \in [0,1]$, $\sum_{i=1}^{d} p_i = 1$

- $H(X) = \sum_{i=1}^{d} p_i \left(-\log(p_i)\right) = \sum_{i=1}^{d} p_i \left(\log\left(\frac{1}{p_i}\right)\right) =$

$\leq \log\left(\sum_{i=1}^{d} p_i \cdot \frac{1}{p_i}\right) = \log(d)$

↑ ($f(x) = \log(x)$ is CONCAVE)



- "$\Rightarrow$": If $p_i = \frac{1}{d} \ \forall i \Rightarrow H(X) = \log(d)$

- "$\Leftarrow$": The inequality is saturated if $f\left(\sum_i p_i x_i\right) = \sum_i p_i f(x_i)$ with $x_i = \frac{1}{p_i}$.
  Now $f(x) = \log(x)$ is strictly CONCAVE $\Rightarrow \sum_i p_i f(x_i) = f\left(\sum_i p_i x_i\right) \Leftrightarrow x_i = \bar{x} \ \forall i$.
  $\Rightarrow \frac{1}{p_i} = \bar{x} \overset{\sum p_i = 1}{\Rightarrow} p_i = \frac{1}{d} \ \forall i$.

- DEF. (JOINT PROB. DISTRIBUTION $P(X,Y)$)
  - $X = \{x_1, \dots, x_d\}$, $Y = \{y_1, \dots, y_D\}$
  $P(x,y)$ joint prob. distribution $\overset{\text{DEF.}}{\Longleftrightarrow}$
  - $P(x,y) \geq 0 \quad \forall x \in X, y \in Y$
  - $\sum_{\substack{x \in X \\ y \in Y}} P(x,y) = 1$

- DEF. (JOINT ENTROPY $H(X,Y)$)
$$H(X,Y) = -\sum_{\substack{x \in X \\ y \in Y}} P(x,y) \log\left(P(x,y)\right)$$

DEF. (MARGINAL DISTRIBUTION $P(x)$ and $P(y)$ given $P(x,y)$).

- Given JOINT PROB. DISTRIB. $P(x,y)$, we define:

$$P(x) := \sum_{y \in Y} P(x,y)$$ (marginal distrib. of X.)

$$P(y) := \sum_{x \in X} P(x,y)$$ (marginal distrib. of Y.)

- $P(x) \geq 0$, $\sum_x P(x) = 1$
- $H(X) = -\sum_{x \in X} P(x) \log(P(x))$

- $P(y) \geq 0$, $\sum_y P(y) = 1$
- $H(Y) = -\sum_{y \in Y} P(y) \log(P(y))$

- TH 1 (SUB-ADDITIVE PROPERTY OF $H(X,Y)$)

$$H(X,Y) \leq H(X) + H(Y)$$

$\left(H(X,Y) = H(X) + H(Y) \iff P(X,Y) = P(X)\cdot P(Y)\right)$ (↑ INDIP. RANDOM VARIABLES)

PROOF:

$$H(X,Y) - H(X) - H(Y) = -\sum_{x,y} P(x,y) \log\left(P(x,y)\right) + \sum_{x} P(x) \log\left(P(x)\right) + \sum_{y} P(y) \log\left(P(y)\right)$$

$$= -\sum_{x,y} P(x,y) \log\left(P(x,y)\right) + \sum_{x,y} P(x,y) \log\left(P(x)\right) + \sum_{x,y} P(x,y) \log\left(P(y)\right)$$

(↑ $\cdot\ P(x) = \sum_y P(x,y)$; $\cdot\ P(y) = \sum_x P(x,y)$)

$$= -\sum_{x,y} P(x,y) \log\left(\frac{P(x,y)}{P(x)\,P(y)}\right) = \sum_{x,y} P(x,y) \log\left(\frac{P(x)\,P(y)}{P(x,y)}\right)$$

$$\leq \log\left(\sum_{x,y} P(x,y) \frac{P(x)\,P(y)}{P(x,y)}\right) = \log(1) = 0$$

(↑ $\cdot\ \log(\cdot)$ CONCAVE; ↑ $\cdot\ \sum_x P(x) = 1$, $\cdot\ \sum_y P(y) = 1$)

- Since $\log(\cdot)$ is strictly CONCAVE $\Rightarrow$ the inequality is saturated when $\frac{P(x)\,P(y)}{P(x,y)} = \text{CONST}\ \ \forall x,y$.

$\iff P(x,y) = P(x)\cdot P(y)$.

↑ Product distribution.

TH ②

$$H(X) \leq H(X,Y) \quad \left("=" \Leftrightarrow P(x,y) = P(x)\,\delta_{y,\bar{y}_0}\right)$$

$$H(Y) \leq H(X,Y) \quad \left("=" \Leftrightarrow P(x,y) = P(y)\,\delta_{x,\bar{x}_0}\right)$$

$$\left(\text{or} \quad \max\left(H(X),H(Y)\right) \leq H(X,Y)\right)$$

PROOF:

$$H(X) - H(X,Y) = -\sum_x \underbrace{P(x)}_{\sum_y P(x,y)} \log\left(P(x)\right) + \sum_{x,y} P(x,y)\log\left(P(x,y)\right)$$

$$= -\sum_{x,y} P(x,y)\log\left(\frac{P(x)}{P(x,y)}\right)$$

$$= \sum_{x,y} P(x,y)\log\left(\frac{P(x,y)}{P(x)}\right) \leq \log\left(\sum_{x,y}\frac{P^2(x,y)}{P(x)}\right)$$

($\log(\cdot)$ CONCAVE)

$$\leq \log\left(\sum_{x,y}\frac{P^2(x,y)}{P(x)}\right) \leq \log\left(\sum_x \frac{\cancel{P^2(x)}}{\cancel{P(x)}}\right) = \log(1) = 0 \quad \Rightarrow \square$$

- $\dfrac{P^2(x,y)}{P(x)} = \dfrac{P(x,y)}{P(x)} \cdot P(x,y) \leq 1 \cdot P(x,y)$
  - $P(x) = \sum_{y'} P(x,y') \Rightarrow \dfrac{P(x,y)}{P(x)} \leq 1$
  - SATURATED $\Leftrightarrow \dfrac{P(x,y)}{P(x)} = 1 \Leftrightarrow P(x,y) = P(x)\cdot\delta_{y,\bar{y}_0}$
- $\log(\cdot)$ monotone

- We'll see that $\max\left(H(X),H(Y)\right) \leq H(X,Y)$ is violated in the "QUANTUM VERSION" due to ENTANGLEMENT.

DEF: (CONDITIONAL PROBABILITY $P(X|Y)$)

$$P(x|y) := \frac{P(x,y)}{P(y)}$$

(Fixed y, $P(x|y)$ is a prob. dist.: $\begin{cases} \cdot\ P(x|y) \geq 0 \\ \cdot\ \sum_x P(x|y) = 1 \end{cases}$; $\sum_x P(x,y) = P(y)$)

↓ This defines a random variable $X|_y$ distributed according to $P(x|y)$.

$$P(y|x) := \frac{P(x,y)}{P(x)}$$

(Fixed x, $P(y|x)$ is a prob. dist.: $\begin{cases} \cdot\ P(y|x) \geq 0 \\ \cdot\ \sum_y P(y|x) = 1 \end{cases}$)

↓ This defines a random variable $Y|_x$

OBS! We have $\sum_x P(x|y)\,P(y) = P(x)$

DEF: (CONDITIONAL ENTROPY $H(X|Y)$)

$$H(X|Y) := \mathop{\mathbb{E}}_y H(X|_y) = \mathop{\mathbb{E}}_y \left(-\sum_x P(x|y) \log\left(P(x|y)\right) \right)$$

COR:

$$H(X|Y) = -\sum_{x,y} P(x,y) \log\left(P(x|y)\right)$$
$$= H(X,Y) - H(Y)$$

PROOF:

- $H(X|Y) = \mathop{\mathbb{E}}_y \left(-\sum_x P(x|y) \log\left(P(x|y)\right) \right)$

$$= -\sum_{x,y} P(x|y)\,P(y) \log\left(P(x|y)\right) =$$

$$= -\sum_{x,y} P(x,y) \log\left(P(x|y)\right)$$

↑ $P(x|y) := \frac{P(x,y)}{P(y)}$

- $H(X|Y) = -\sum_{x,y} P(x,y) \log\left(P(x|y)\right) = -\sum_{x,y} P(x,y) \log\left(\frac{P(x,y)}{P(y)}\right) =$

$$= -\sum_{x,y} P(x,y) \log\left(P(x,y)\right) + \sum_{y,x} \underbrace{P(x,y)}_{P(y)} \log\left(P(y)\right) = H(X,Y) - H(Y)$$

OBS. $H(X|Y) \geq 0$

PROOF:

$$H(X|Y) = H(X,Y) - H(Y) \geq 0$$

↑ THEOREM (2)

(This inequality does not hold for the Quantum version of $H(X|Y)$ because TH.(2) is violated.)

DEF. (MUTUAL INFORMATION)

$$I(X:Y) := H(X) + H(Y) - H(X,Y)$$

OBS.

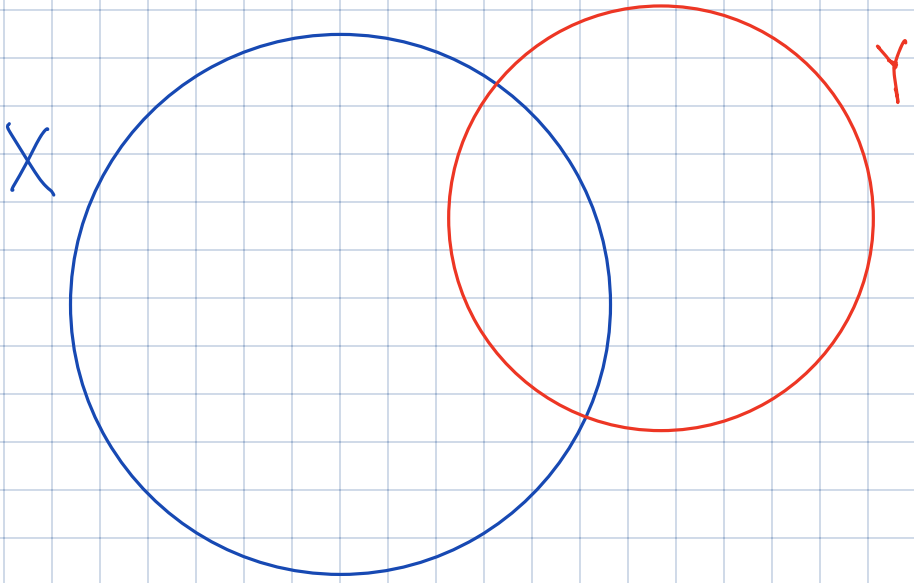
- $I(X:Y) \geq 0$ ↑ (SUB-ADD. (TH.(2)))
- $I(X:Y) = I(Y:X)$ ↑ $H(X,Y) = H(Y,X)$
- $I(X:Y) = H(X,Y) - H(X|Y) - H(Y|X) = H(Y) - H(Y|X)$

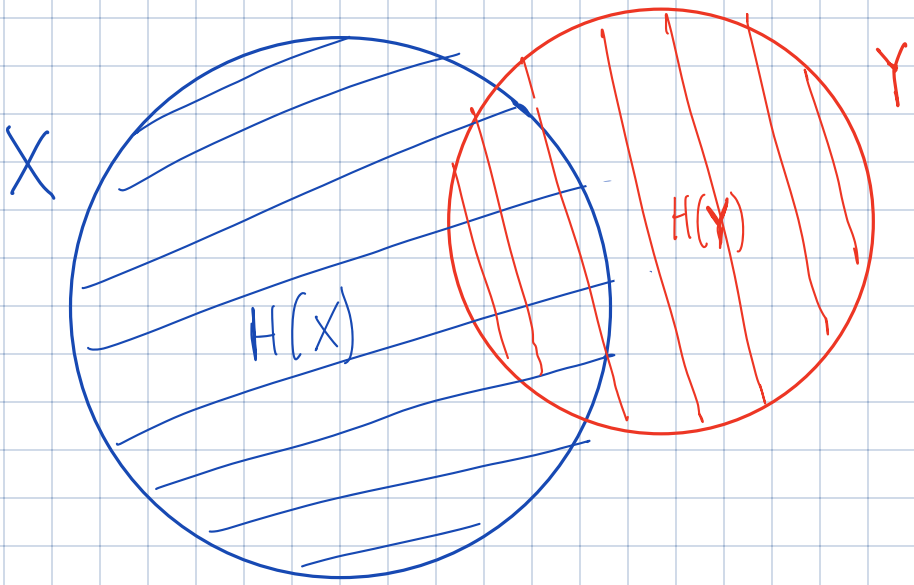
  ↑ $I(X:Y) = \big(H(X,Y) - H(X|Y)\big) + \big(H(X,Y) - H(Y|X)\big) - H(X,Y)$

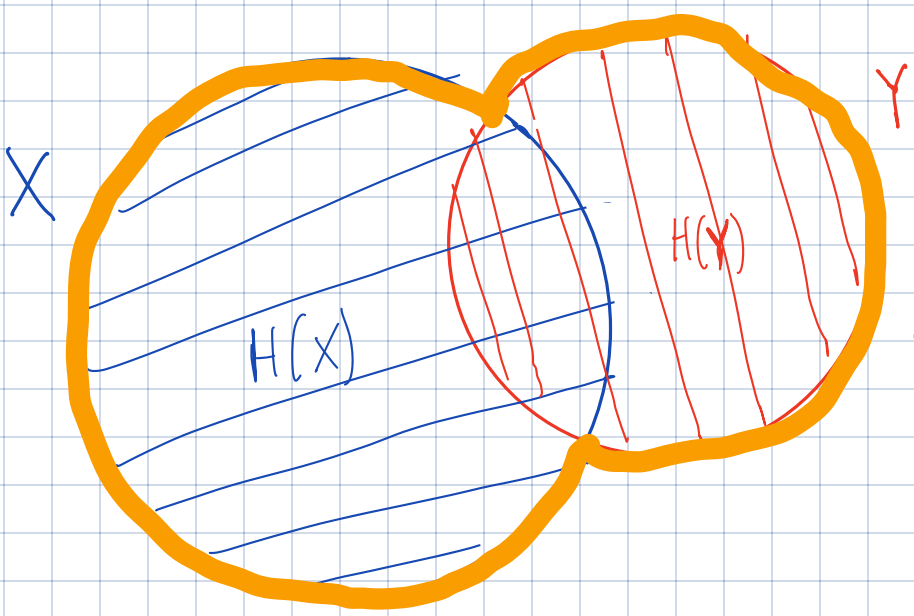
  ↑
  - $H(X|Y) = H(X,Y) - H(Y)$
  - $H(Y|X) = H(X,Y) - H(X)$
- $I(X:Y) \leq \min\big(H(X), H(Y)\big)$

  ↑ $I(X:Y) = H(X) + H(Y) - H(X,Y) \leq H(X) + H(Y) - \max\big(H(X), H(Y)\big) = \min\big(H(X), H(Y)\big)$

GRAPHICAL INTERPRETATION

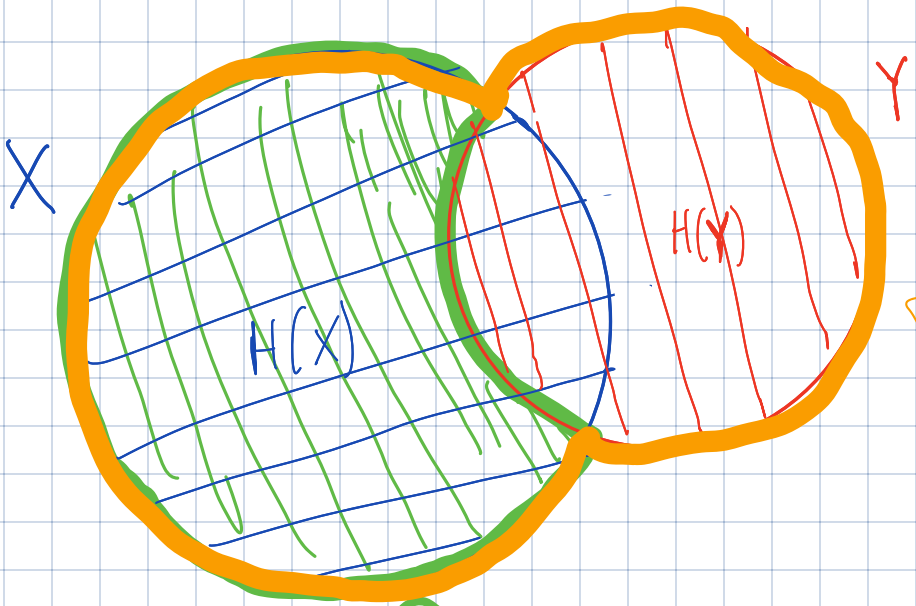


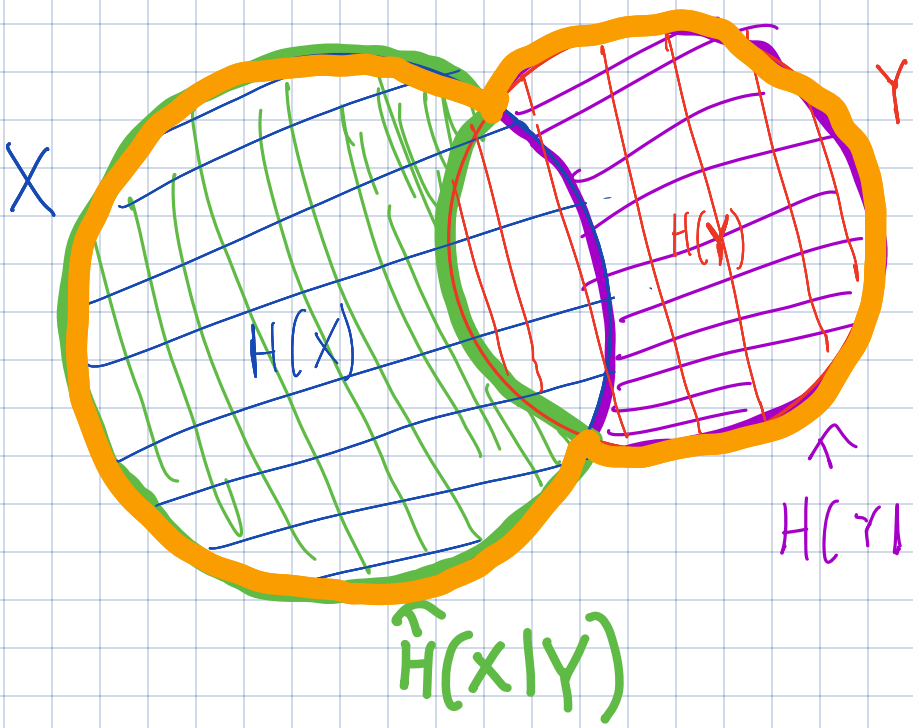


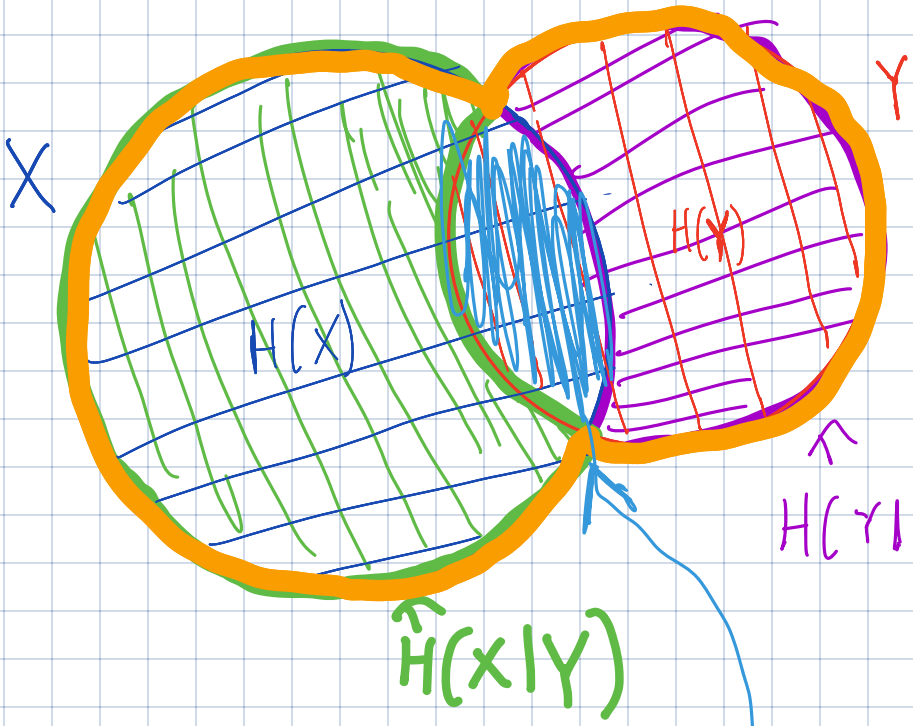


$\leftarrow H(X,Y)$

- $H(X,Y) \leq H(X) + H(Y)$
- $H(X,Y) \geq \max(H(X), H(Y))$
- They are obvious here!


X
Y
H(X)
H(Y)
H(X,Y)
H(X|Y) = H(X,Y) - H(Y)


X
Y
H(X)
H(Y)
H(X,Y)
H(Y|X) = H(X,Y) - H(X)
H(X|Y)



$$I(X:Y) = H(X,Y) - H(X|Y) - H(Y|X)$$
$$= H(X) - H(X|Y)$$
$$= H(Y) - H(Y|X)$$

- DEF: (RELATIVE ENTROPY)

- $p(x) \geq 0$; $\sum_x p(x) = 1$. $q(x) \geq 0$; $\sum_x q(x) = 1$.

$$H(p(x) \| q(x)) := \sum_x p(x) \log_2\left(\frac{p(x)}{q(x)}\right) =$$

$$= -H(p) - \sum_x p(x) \log_2(q(x))$$

• OBS.

$$H(p(x) \| q(x)) \geq 0 \qquad \left(H(p(x) \| q(x)) = 0 \iff p(x) = q(x) \ \forall x \right)$$

PROOF:

$$H(p(x) \| q(x)) := \sum_x p(x) \log_2\left(\frac{p(x)}{q(x)}\right) \geq \sum_x \frac{p(x)}{\ln(2)} \left(1 - \frac{q(x)}{p(x)}\right)$$

($\uparrow$)

- $e^x \geq 1 + x$
- $\Rightarrow x \geq \ln(1+x)$
- $\Rightarrow y - 1 \geq \ln(y) = \log_2(y) \ln(2)$
- $\Rightarrow -\log_2(y) \geq \frac{1-y}{\ln(2)}$
- $\log_2\left(\frac{p}{q}\right) = -\log_2\left(\frac{q}{p}\right)$

$$= \frac{1}{\ln(2)}(1 - 1) = 0$$

• The inequality $\log_2(y) \geq \frac{1}{\ln(2)}(1 - y)$ is saturated $\iff y = 1 \iff p(x) = q(x)$

OBS:

$$\bullet \quad H(p(x,y) \| p(x) \cdot p(y)) \underset{\text{DEF}}{=} -H(X,Y) + H(X) + H(Y) \underset{\text{OBS.}}{\geq} 0$$

$$\Rightarrow \quad H(X,Y) \leq H(X) + H(Y)$$

$$\bullet \quad H(p(x) \| q(x)) \underset{q(x) = \frac{1}{d} \ \forall x}{=} -H(X) - \underbrace{\sum_x p(x)}_{1} \log\left(\frac{1}{d}\right) \underset{\text{OBS}}{\geq} 0 \Rightarrow H(X) \leq \log(d)$$